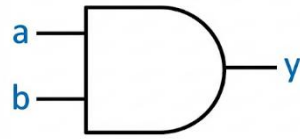


AND Gate

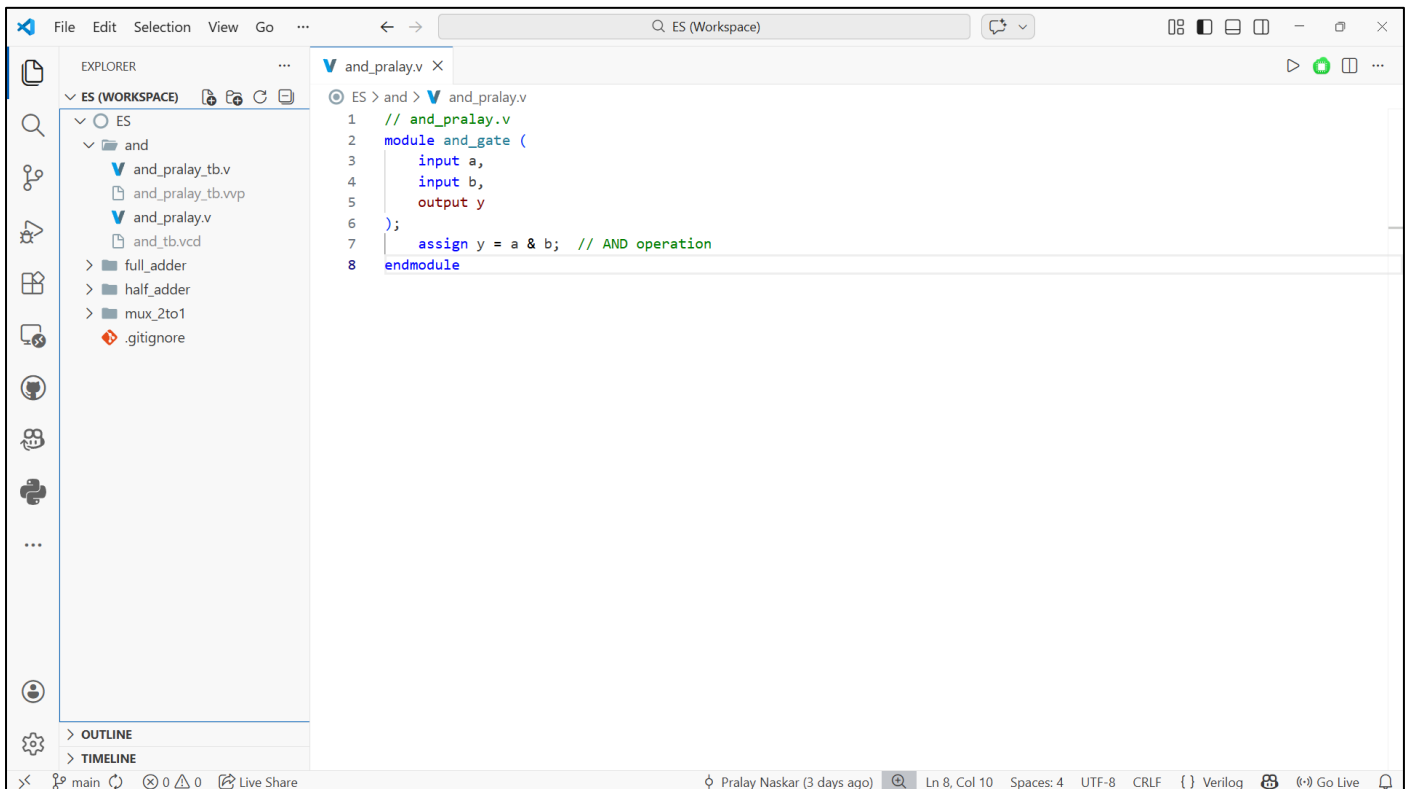
AND Gate: Symbol and Truth Table



Input a	Input b	Output y
0	0	0
0	1	0
1	0	0
1	1	1

Verilog Code

```
// and_pralay.v
module and_gate (
    input a,
    input b,
    output y
);
    assign y = a & b; // AND operation
endmodule
```



```

// and_pralay_tb.v
`timescale 1ns/1ps
module and_gate_tb;

    // Testbench signals
    reg a;
    reg b;
    wire y;

    // Instantiate the AND gate
    and_gate uut (
        .a(a),
        .b(b),
        .y(y)
    );

    // Stimulus
    initial begin
        $dumpfile("and_tb.vcd");
        $dumpvars(0, and_gate_tb);

        $display("A B | Y");
        $display("-----");

        a = 0; b = 0; #10;
        $display("%b %b | %b", a, b, y);

        a = 0; b = 1; #10;
        $display("%b %b | %b", a, b, y);

        a = 1; b = 0; #10;
        $display("%b %b | %b", a, b, y);

        a = 1; b = 1; #10;
        $display("%b %b | %b", a, b, y);

        $finish;
    end

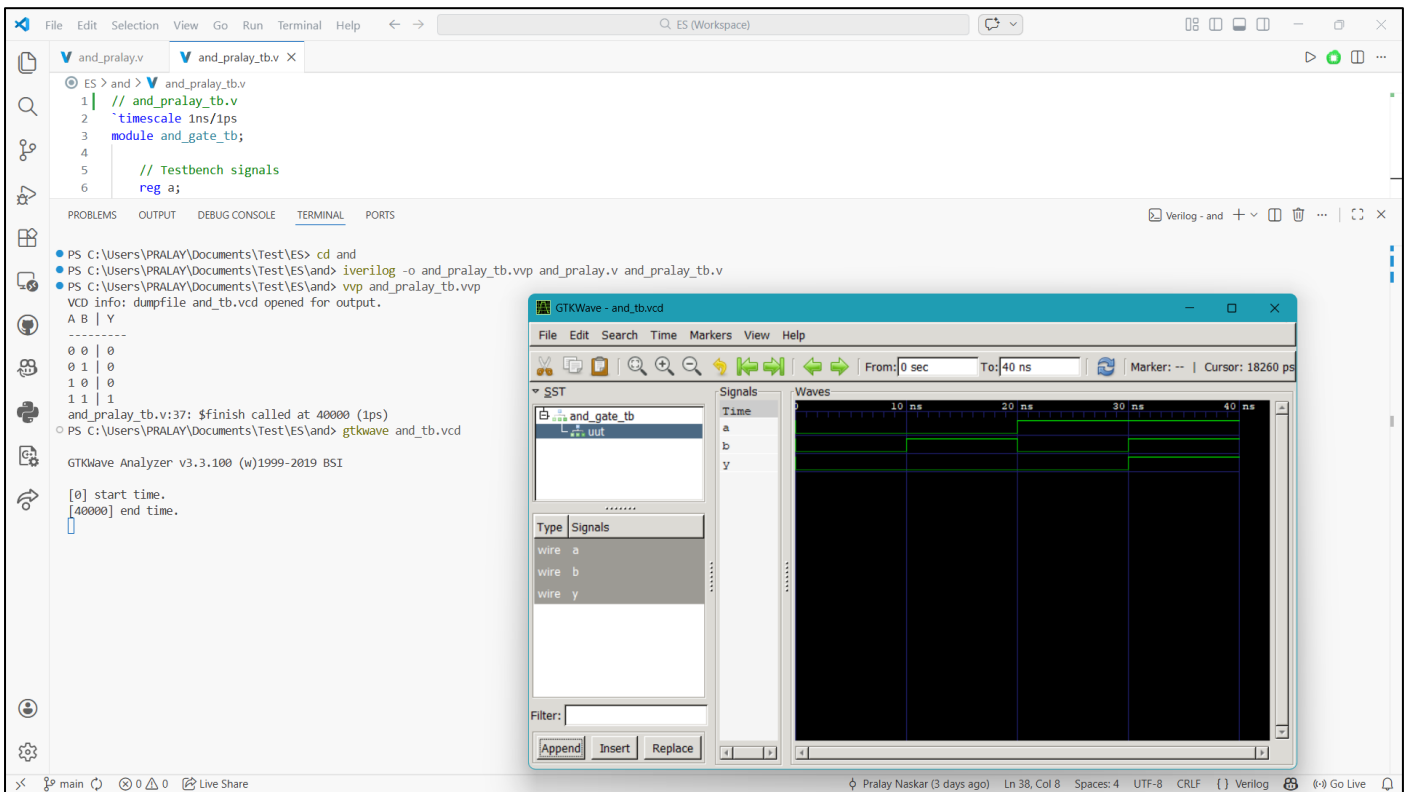
endmodule

```

The screenshot shows the VS Code editor with a workspace named 'ES'. The Explorer panel on the left shows a project structure with files like 'and', 'and_pralay_tb.v', 'and_pralay_tb.vvp', 'and_pralay.v', 'and_tb.vcd', 'full_adder', 'half_adder', 'mux_2to1', and '.gitignore'. The main editor window displays the Verilog code for 'and_pralay_tb.v'.

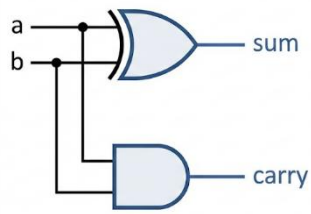
```
1 | // and_pralay_tb.v
2 | `timescale 1ns/1ps
3 | module and_gate_tb;
4 |
5 |     // Testbench signals
6 |     reg a;
7 |     reg b;
8 |     wire y;
9 |
10 |    // Instantiate the AND gate
11 |    and_gate uut (
12 |        .a(a),
13 |        .b(b),
14 |        .y(y)
15 |    );
16 |
17 |    // Stimulus
18 |    initial begin
19 |        $dumpfile("and_tb.vcd");
20 |        $dumpvars(0, and_gate_tb);
21 |
22 |        $display("A B | Y");
23 |        $display("-----");
24 |
25 |        a = 0; b = 0; #10;
26 |        $display("%b %b | %b", a, b, y);
27 |
28 |        a = 0; b = 1; #10;
29 |        $display("%b %b | %b", a, b, y);
30 |
31 |        a = 1; b = 0; #10;
32 |        $display("%b %b | %b", a, b, y);
33 |
34 |        a = 1; b = 1; #10;
35 |        $display("%b %b | %b", a, b, y);
36 |
37 |        $finish;
38 |    end
39 |
40 | endmodule
```

OUTPUT



Half-Adder Gate

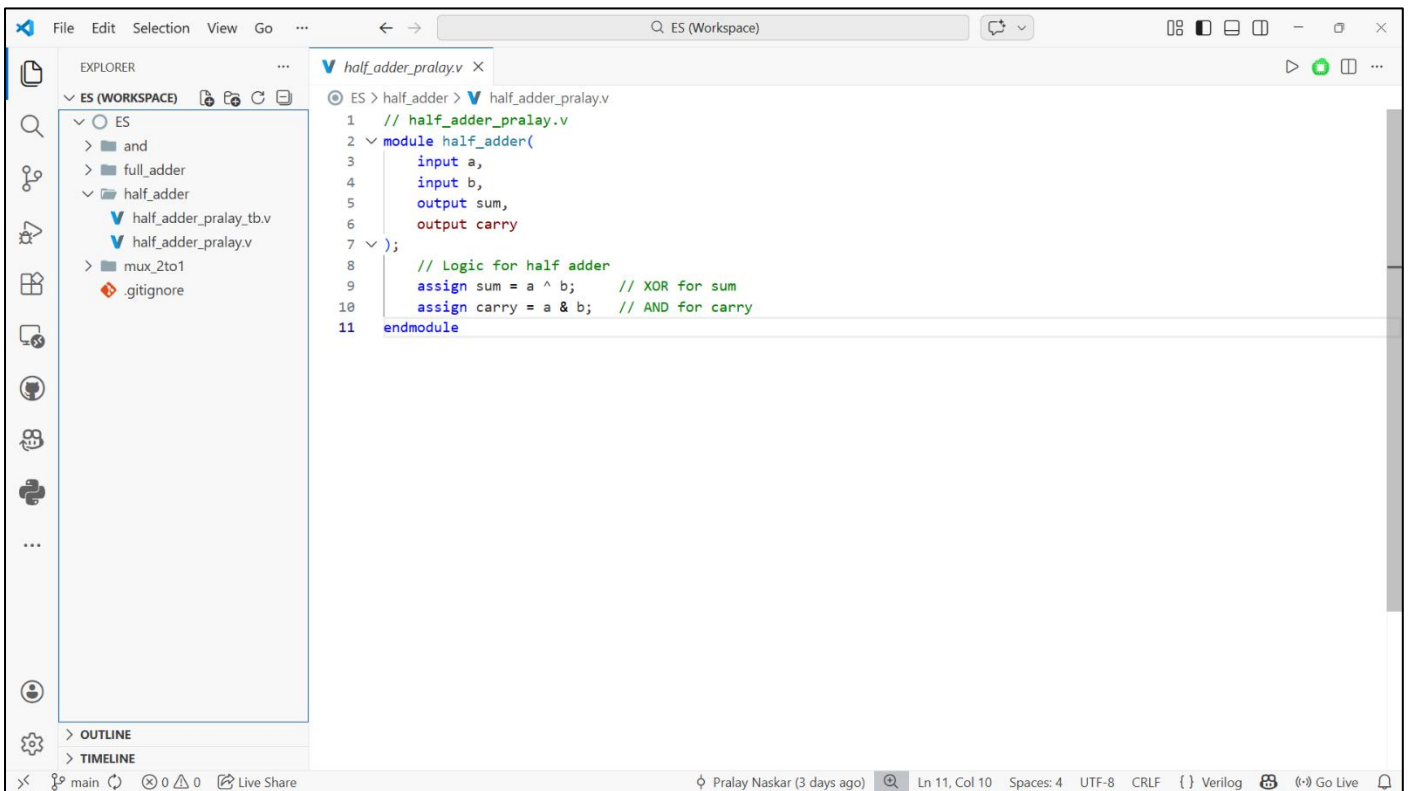
Half-Adder Circuit and Truth Table



Input		Output	
a	b	sum	carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Verilog Code

```
// half_adder_pralay.v
module half_adder(
    input a,
    input b,
    output sum,
    output carry
);
    // Logic for half adder
    assign sum = a ^ b;      // XOR for sum
    assign carry = a & b;    // AND for carry
endmodule
```



```

// half_adder_pralay_tb.v
`timescale 1ns/1ps
module half_adder_tb;

    reg a;
    reg b;
    wire sum;
    wire carry;

    // Instantiate the half adder
    half_adder uut (
        .a(a),
        .b(b),
        .sum(sum),
        .carry(carry)
    );

    // Test sequence
    initial begin
        $dumpfile("half_adder.vcd");    // VCD file for GTKWave
        $dumpvars(0, half_adder_tb);

        $display("A B | Sum Carry");
        $display("-----");

        a = 0; b = 0; #10;
        $display("%b %b | %b %b", a, b, sum, carry);

        a = 0; b = 1; #10;
        $display("%b %b | %b %b", a, b, sum, carry);

        a = 1; b = 0; #10;
        $display("%b %b | %b %b", a, b, sum, carry);

        a = 1; b = 1; #10;
        $display("%b %b | %b %b", a, b, sum, carry);

        $finish;
    end

endmodule

```

```
1 // half_adder_pralay.tb.v
2 `timescale 1ns/1ps
3 module half_adder_tb;
4
5     reg a;
6     reg b;
7     wire sum;
8     wire carry;
9
10    // Instantiate the half adder
11    half_adder uut (
12        .a(a),
13        .b(b),
14        .sum(sum),
15        .carry(carry)
16    );
17
18    // Test sequence
19    initial begin
20        $dumpfile("half_adder.vcd"); // VCD file for GTKWave
21        $dumpvars(0, half_adder_tb);
22
23        $display("A B | Sum Carry");
24        $display("-----");
25
26        a = 0; b = 0; #10;
27        $display("%b %b | %b %b", a, b, sum, carry);
28
29        a = 0; b = 1; #10;
30        $display("%b %b | %b %b", a, b, sum, carry);
31
32        a = 1; b = 0; #10;
33        $display("%b %b | %b %b", a, b, sum, carry);
34
35        a = 1; b = 1; #10;
36        $display("%b %b | %b %b", a, b, sum, carry);
37
38        $finish;
39    end
40
41 endmodule
```

OUTPUT

Terminal Output:

```
PS C:\Users\PRALAY\Documents\Test\ES> cd half_adder
PS C:\Users\PRALAY\Documents\Test\ES\half_adder> iverilog -o half_adder_pralay_tb.vvp half_adder_pralay.v half_adder_pralay_tb.v
PS C:\Users\PRALAY\Documents\Test\ES\half_adder> vvp half_adder_pralay_tb.vvp
VCD info: dumpfile half_adder.vcd opened for output.
A B | Sum Carry
0 0 | 0 0
0 1 | 1 0
1 0 | 1 0
1 1 | 0 1
half_adder_pralay_tb.v:38: $finish called at 40000 (1ps)
PS C:\Users\PRALAY\Documents\Test\ES\half_adder> gtkwave half_adder.vcd
GTKWave Analyzer v3.3.100 (w)1999-2019 BSI
[0] start time.
[40000] end time.
```

GTKWave - half_adder.vcd

Signals:

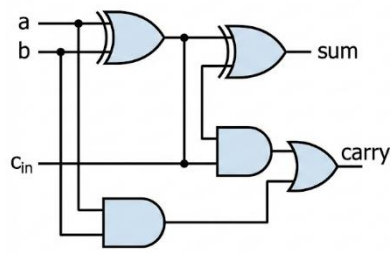
- half_adder_tb
- uut

Waves:

- a
- b
- carry
- sum

Full-Adder

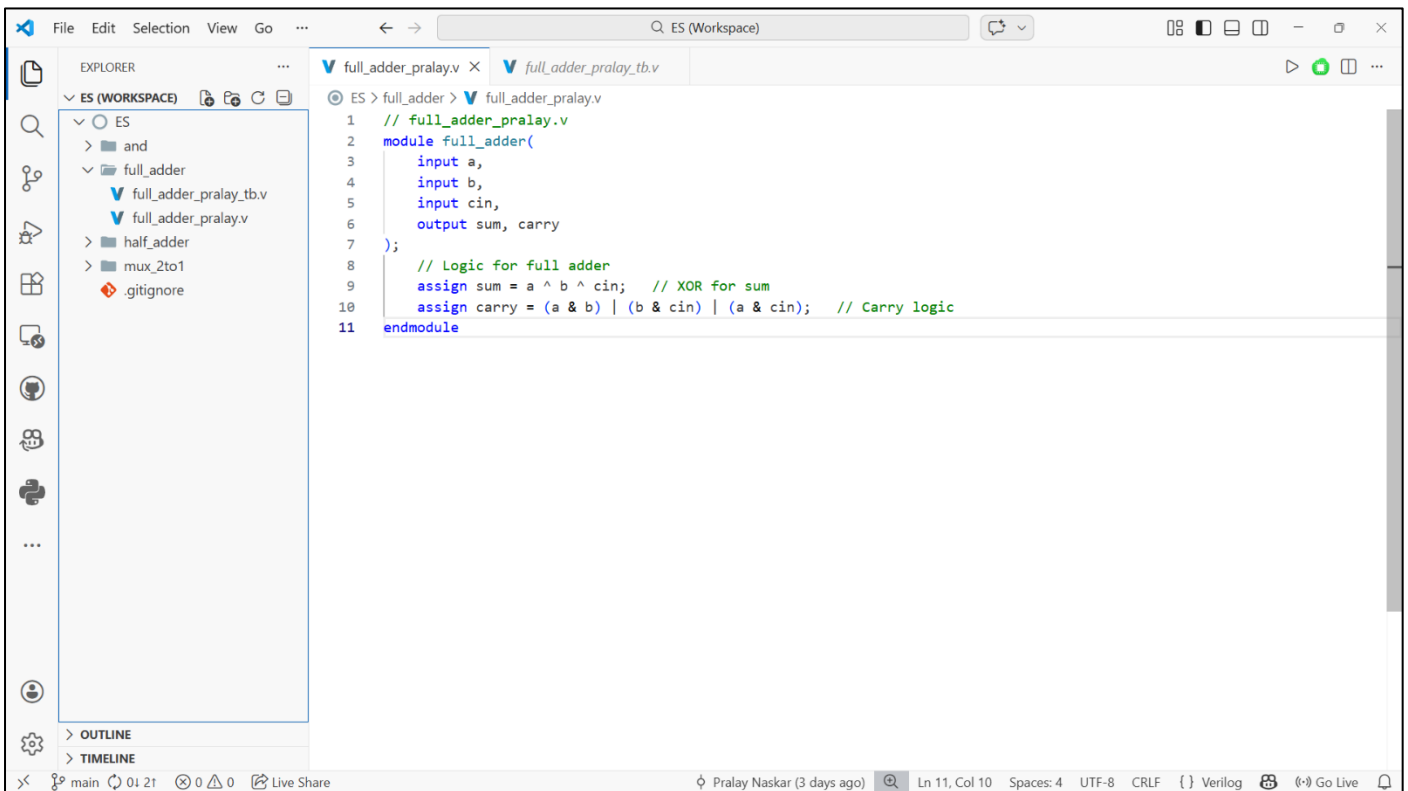
Full-Adder Circuit and Truth Table



Input			Output	
a	b	c _{in}	sum	carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Verilog Code

```
// full_adder_pralay.v
module full_adder(
    input a,
    input b,
    input cin,
    output sum, carry
);
    // Logic for full adder
    assign sum = a ^ b ^ cin;    // XOR for sum
    assign carry = (a & b) | (b & cin) | (a & cin);    // Carry Logic
endmodule
```



```

// full_adder_pralay_tb.v
`timescale 1ns/1ps
module full_adder_tb;

    reg a;
    reg b;
    reg cin;
    wire sum;
    wire carry;

    // Instantiate the full adder
    full_adder uut (
        .a(a),
        .b(b),
        .cin(cin),
        .sum(sum),
        .carry(carry)
    );

    // Test sequence
    initial begin
        $dumpfile("full_adder.vcd"); // VCD file for GTKWave
        $dumpvars(0, full_adder_tb);

        $display("A B Cin | Sum Carry");
        $display("-----");

        a = 0; b = 0; cin = 0; #10;
        $display("%b %b %b | %b %b", a, b, cin, sum, carry);

        a = 0; b = 0; cin = 1; #10;
        $display("%b %b %b | %b %b", a, b, cin, sum, carry);

        a = 0; b = 1; cin = 0; #10;
        $display("%b %b %b | %b %b", a, b, cin, sum, carry);

        a = 0; b = 1; cin = 1; #10;
        $display("%b %b %b | %b %b", a, b, cin, sum, carry);

        a = 1; b = 0; cin = 0; #10;
        $display("%b %b %b | %b %b", a, b, cin, sum, carry);

        a = 1; b = 0; cin = 1; #10;
        $display("%b %b %b | %b %b", a, b, cin, sum, carry);

        a = 1; b = 1; cin = 0; #10;
        $display("%b %b %b | %b %b", a, b, cin, sum, carry);

        a = 1; b = 1; cin = 1; #10;
        $display("%b %b %b | %b %b", a, b, cin, sum, carry);

        $finish;
    end

endmodule

```



```
File Edit Selection View Go Run Terminal Help
Q ES (Workspace)

EXPLORER
ES (WORKSPACE)
  > full_adder
    full_adder
    full_adder_tb.v
    full_adder_pralay.v
    full_adder_pralay_tb.v
    mux_2to1
    qifignore

full_adder_pralay.v
1 // full_adder_pralay_tb.v
2 `timescale 1ns/1ps
3 module full_adder_tb;
4
5     reg a;
6     reg b;
7     reg cin;
8     wire sum;
9     wire carry;
10
11 // Instantiate the full adder
12 full_adder uut (
13     .a(a),
14     .b(b),
15     .cin(cin),
16     .sum(sum),
17     .carry(carry)
18 );
19
20 // Test sequence
21 initial begin
22     $dumpfile("full_adder.vcd"); // VCD file for GTKWave
23     $dumpvars(0, full_adder_tb);
24
25     $display("A B Cin | Sum Carry");
26     $display("-----");
27
28     a = 0; b = 0; cin = 0; #10;
29     $display("%0 %0 %0 | %0 %0", a, b, cin, sum, carry);
30
31     a = 0; b = 0; cin = 1; #10;
32     $display("%0 %0 %0 | %0 %0", a, b, cin, sum, carry);
33
34     a = 0; b = 1; cin = 0; #10;
35     $display("%0 %0 %0 | %0 %0", a, b, cin, sum, carry);
36
37     a = 0; b = 1; cin = 1; #10;
38     $display("%0 %0 %0 | %0 %0", a, b, cin, sum, carry);
39
40     a = 1; b = 0; cin = 0; #10;
41     $display("%0 %0 %0 | %0 %0", a, b, cin, sum, carry);
42
43     a = 1; b = 0; cin = 1; #10;
44     $display("%0 %0 %0 | %0 %0", a, b, cin, sum, carry);
45
46     a = 1; b = 1; cin = 0; #10;
47     $display("%0 %0 %0 | %0 %0", a, b, cin, sum, carry);
48
49     a = 1; b = 1; cin = 1; #10;
50     $display("%0 %0 %0 | %0 %0", a, b, cin, sum, carry);
51
52     $finish;
53 end
54
55 endmodule
```

OUTPUT

File Edit Selection View Go Run Terminal Help

Q ES (Workspace)

full_adder_pralay.v full_adder_pralay_tb.v

```
1 // full_adder_pralay_tb.v
2 `timescale 1ns/1ps
3 module full_adder_tb;
4
5     reg a;
6     reg b;
7     reg cin;
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\PRALAY\Documents\Test\ES> cd full_adder

PS C:\Users\PRALAY\Documents\Test\ES\full_adder> iverilog -o full_adder_pralay_tb.vvp full_adder_pralay.v full_adder_pralay_tb.v

PS C:\Users\PRALAY\Documents\Test\ES\full_adder> vvp full_adder_pralay_tb.vvp

VCD info: dumpfile full_adder.vcd opened for output.

A B Cin | Sum Carry

```
0 0 0 | 0 0
0 0 1 | 1 0
0 1 0 | 1 0
0 1 1 | 0 1
1 0 0 | 1 0
1 0 1 | 0 1
1 1 0 | 0 1
1 1 1 | 1 1
```

full_adder_pralay_tb.v:52: \$finish called at 80000 (1ps)

PS C:\Users\PRALAY\Documents\Test\ES\full_adder> gtkwave full_adder.vcd

GTKWave Analyzer v3.3.100 (w)1999-2019 BSI

[0] start time.

[80000] end time.

GTKWave - full_adder.vcd

File Edit Search Time Markers View Help

From: 0 sec To: 80 ns Marker: -- Cursor: 900 ps

SST

full_adder_tb

uut

Signals

Time

a

b

carry

cin

sum

Type Signals

wire a

wire b

wire carry

wire cin

wire sum

Filter:

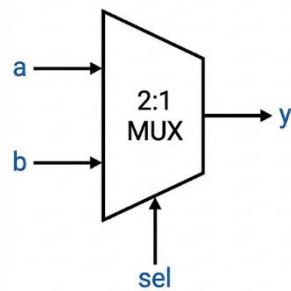
Append Insert Replace

Waves

0

2 Input Multiplexer

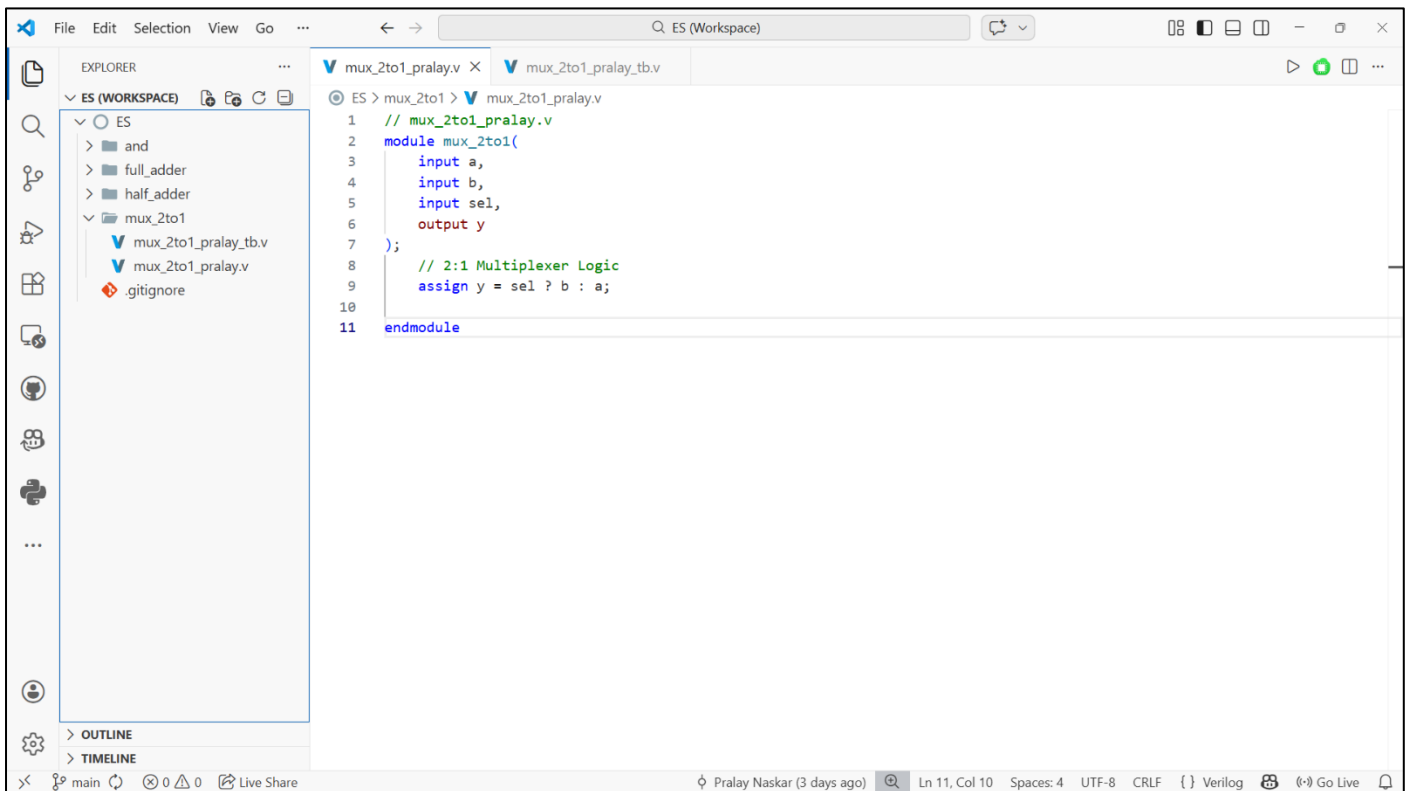
2-Input Multiplexer (MUX) Circuit and Truth Table



Input sel	Input a	Input b	Output y
0	X	X	a
1	X	X	b
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

Verilog Code

```
// mux_2to1_pralay.v
module mux_2to1(
    input a,
    input b,
    input sel,
    output y
);
    // 2:1 Multiplexer Logic
    assign y = sel ? b : a;
endmodule
```



```

// mux_2to1_pralay_tb.v
`timescale 1ns/1ps
module mux_2to1_tb;

    reg a;
    reg b;
    reg sel;
    wire y;

    // Instantiate the 2:1 MUX
    mux_2to1 uut (
        .a(a),
        .b(b),
        .sel(sel),
        .y(y)
    );

    // Test sequence
    initial begin
        $dumpfile("mux_2to1.vcd"); // VCD file for GTKWave
        $dumpvars(0, mux_2to1_tb);

        $display("A B Sel | Y");
        $display("-----");

        a = 0; b = 0; sel = 0; #10;
        $display("%b %b %b | %b", a, b, sel, y);

        a = 0; b = 1; sel = 0; #10;
        $display("%b %b %b | %b", a, b, sel, y);

        a = 1; b = 0; sel = 0; #10;
        $display("%b %b %b | %b", a, b, sel, y);

        a = 1; b = 1; sel = 0; #10;
        $display("%b %b %b | %b", a, b, sel, y);

        a = 0; b = 0; sel = 1; #10;
        $display("%b %b %b | %b", a, b, sel, y);

        a = 0; b = 1; sel = 1; #10;
        $display("%b %b %b | %b", a, b, sel, y);

        a = 1; b = 0; sel = 1; #10;
        $display("%b %b %b | %b", a, b, sel, y);

        a = 1; b = 1; sel = 1; #10;
        $display("%b %b %b | %b", a, b, sel, y);

        $finish;
    end
endmodule

```

```
File Edit Selection View Go Run Terminal Help
Q ES (Workspace)

EXPLORER
ES (WORKSPACE)
  > ES
  > full_adder
  > full_adder
  > half_adder
  > half_adder
  > mux_2to1
    mux_2to1.pralay.v
    mux_2to1.pralay.tb.v
    mux_2to1.pralay.vvp
    mux_2to1.vcd
    .gitignore

mux_2to1.pralay.tb.v
1 // mux_2to1.tb.v
2 `timescale 1ns/1ps
3 module mux_2to1_tb;
4
5     reg a;
6     reg b;
7     reg sel;
8     wire y;
9
10 // Instantiate the 2:1 MUX
11 mux_2to1 uut (
12     .a(a),
13     .b(b),
14     .sel(sel),
15     .y(y)
16 );
17
18 // Test sequence for all possible input combinations
19 initial begin
20
21     $dumpfile("mux_2to1.vcd");
22     $dumpvars(0, mux_2to1_tb);
23
24     $display("A B Sel | Y");
25     $display("-----");
26
27     a = 0; b = 0; sel = 0; #10;
28     $display("%0 %0 %0 | %0", a, b, sel, y);
29
30     a = 0; b = 1; sel = 0; #10;
31     $display("%0 %0 %0 | %0", a, b, sel, y);
32
33     a = 1; b = 0; sel = 0; #10;
34     $display("%0 %0 %0 | %0", a, b, sel, y);
35
36     a = 1; b = 1; sel = 0; #10;
37     $display("%0 %0 %0 | %0", a, b, sel, y);
38
39     a = 0; b = 0; sel = 1; #10;
40     $display("%0 %0 %0 | %0", a, b, sel, y);
41
42     a = 0; b = 1; sel = 1; #10;
43     $display("%0 %0 %0 | %0", a, b, sel, y);
44
45     a = 1; b = 0; sel = 1; #10;
46     $display("%0 %0 %0 | %0", a, b, sel, y);
47
48     a = 1; b = 1; sel = 1; #10;
49     $display("%0 %0 %0 | %0", a, b, sel, y);
50
51     $finish;
52
53 end
54
55 endmodule

> OUTLINE
> TIMELINE
main 01:11 0 0 0 Live Share
Pralay Naskar (4 days ago) Ln 30, Col 36 Spaces: 4 UTF-8 CRLF Verilog Go Live
```

OUTPUT

```
File Edit Selection View Go Run Terminal Help
Q ES (Workspace)

mux_2to1.pralay.v mux_2to1.pralay.tb.v
ES > mux_2to1 > mux_2to1.pralay.tb.v
1 // mux_2to1.tb.v
2 `timescale 1ns/1ps
3 module mux_2to1_tb;
4
5     reg a;
6     reg b;
7     reg sel;
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